

PASSO ROLLE, Italy

WMO: 160210

Lat: 46.297N

Lon: 11.787E

Elev: 2006

StdP: 79.44

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.4	-12.7	-29.8	0.3	-5.4	-26.7	0.4	-6.1	12.4	-4.2	10.4	-4.5	2.3	330	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	19.2	12.7	18.0	12.4	16.8	11.9	14.7	17.2	13.7	16.3	12.9	15.4	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	12.6	15.9	12.8	11.8	15.1	11.8	11.1	14.3	48.6	17.7	45.5	16.3	42.9	15.4	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.8	8.0	6.9		-15.8	21.6	2.6	1.2	-17.7	22.4	-19.2	23.1	-20.7	23.7	-22.6	24.5	
(5)			DB	-16.8	16.8	2.4	1.2	-18.5	17.7	-19.9	18.4	-21.3	19.1	-23.0	20.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-4.3	-4.2	-1.9	1.4	5.5	10.1	12.4	11.9	8.2	4.9	0.5	-3.0		
	DBStd	6.93	3.64	4.37	4.00	3.17	3.37	3.52	2.66	2.80	3.05	3.50	3.98	4.05		
	HDD10.0	2596	444	398	369	258	143	42	9	14	68	161	286	403		
	HDD18.3	5415	702	632	627	508	397	247	184	200	303	418	536	661		
	CDD10.0	223	0	0	0	0	5	46	84	72	15	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.7	2.8	2.5	2.8	2.7	2.9	2.9	2.9	2.9	2.6	2.5	2.6	2.7		
(15) Precipitation	PrecAvg	1089	52	53	66	91	115	115	106	104	94	133	99	52		
	PrecMax	1511	175	474	148	200	229	202	195	187	274	374	328	142		
	PrecMin	678	0	1	17	47	54	52	21	52	6	2	0	13		
	PrecStd	202	39	92	31	35	50	41	35	33	60	86	86	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	14.8	8.2	10.9	16.8	20.2	21.0	20.8	17.1	13.8	11.0	7.0		
		MCWB	-0.8	-3.9	2.2	5.5	8.0	12.1	13.7	12.3	11.6	5.8	4.0	-0.8		
	2%	DB	3.5	5.2	6.4	8.8	14.1	18.7	19.5	18.9	15.1	12.0	8.3	4.9		
		MCWB	-1.2	-1.1	1.2	4.1	8.6	12.0	13.3	12.6	10.4	5.6	3.0	-1.3		
	5%	DB	1.9	3.0	4.9	7.1	12.2	17.0	18.1	17.2	13.8	10.2	6.7	3.3		
		MCWB	-2.2	-1.3	0.5	3.2	7.6	11.4	12.8	12.4	9.9	5.7	2.1	-1.8		
	10%	DB	0.2	1.0	3.1	5.8	10.4	15.2	16.8	16.0	12.2	9.1	5.0	1.9		
		MCWB	-2.9	-2.8	-0.1	2.6	6.7	10.8	12.1	11.9	9.1	5.3	1.1	-1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.1	3.5	6.9	11.1	15.2	16.5	15.7	13.5	9.5	6.0	1.6		
		MCDB	3.9	3.7	6.0	9.3	13.8	17.6	18.8	17.5	15.4	11.0	8.8	3.5		
	2%	WB	-0.4	-0.2	2.2	5.2	9.5	13.8	15.0	14.4	11.9	8.3	4.3	0.6		
		MCDB	1.7	2.3	4.6	7.6	12.4	16.6	17.5	16.8	13.7	9.8	7.0	2.3		
	5%	WB	-1.3	-1.4	1.2	4.0	8.5	12.6	14.0	13.4	10.7	7.3	3.2	-0.4		
		MCDB	0.6	1.0	3.6	6.1	11.4	15.2	16.4	16.0	12.6	9.0	5.1	1.3		
	10%	WB	-2.1	-2.3	0.2	2.9	7.4	11.6	13.0	12.6	9.8	6.4	2.3	-1.1		
		MCDB	-0.4	-0.2	2.4	4.7	10.0	14.3	15.5	15.1	11.6	8.1	3.9	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.9	5.9	5.9	6.5	6.9	7.1	6.6	5.9	5.2	4.4	4.4		
		MCDBR	5.8	9.2	6.7	7.0	8.1	8.2	7.9	7.3	6.7	6.2	5.5	5.4		
	5% WB	MCWBR	4.2	4.1	4.8	4.5	5.4	5.8	5.8	5.2	4.9	4.5	3.7	3.5		
		MCWBR	4.6	5.4	6.2	6.4	7.4	7.6	7.4	6.9	6.2	5.1	4.5	4.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.246	0.271	0.317	0.366	0.374	0.388	0.376	0.370	0.347	0.325	0.281	0.248		
		taud	2.429	2.379	2.322	2.241	2.263	2.259	2.294	2.328	2.375	2.417	2.496	2.490		
	Ebn at Noon	Ebn at Noon	880	915	910	889	891	877	883	875	866	835	824	841		
		Edh at Noon	67	87	108	129	131	132	126	117	102	84	64	57		
(44) All-Sky Solar Radiation	RadAvg	1.56	2.46	3.70	4.48	5.08	5.79	5.88	5.10	3.89	2.55	1.53	1.31			
	RadStd	0.18	0.31	0.46	0.59	0.55	0.41	0.40	0.44	0.37	0.33	0.25	0.16			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)		+0.40	N/A	N/A	+0.63	+0.39	+0.34	N/A	-118	+77	+27	

Nomenclature: See separate page